

Loewy lengths from ideal zero-product probabilities

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Abstract

For a finite commutative principal ideal ring, let ζ_k be the probability that the product of k independently and uniformly chosen ideals is zero. We prove that the sequence $(\zeta_k)_{k \geq 2}$ determines the multiset of Loewy lengths of its nonzero chain-ring factors, including all multiplicities. This proves Conjecture 4.11 of Chakraborty and Patra in arXiv:2608.14618v1. The known chain-ring counting formula expresses each complementary probability as a polynomial times an exponential. After common factors are cancelled, the least remaining length dominates the logarithmic probability ratio. We also give a rational error bound that certifies separation for any fixed pair of distinct length multisets.

1 Statement and context

All rings in the theorem are nonzero, finite, commutative, and have an identity. Write $\mathcal{I}(R)$ for the set of ideals of R , including R and (0) . For each integer $k \geq 2$, define

$$\zeta_k(R) = \frac{\#\{(I_1, \dots, I_k) \in \mathcal{I}(R)^k : I_1 \cdots I_k = (0)\}}{|\mathcal{I}(R)|^k}. \quad (1)$$

Thus the distribution is uniform on ideals, without weighting by their cardinalities. A finite chain ring is a ring whose ideals are totally ordered by inclusion. If M is its maximal ideal, its Loewy length is the least positive integer n with $M^n = (0)$. In particular, fields have Loewy length 1.

Theorem 1. *Let R and S be finite commutative principal ideal rings with decompositions into nonzero finite chain rings*

$$R \simeq \prod_{i=1}^m R_i, \quad S \simeq \prod_{j=1}^{\ell} S_j.$$

Let n_i and t_j denote the respective positive Loewy lengths. If $\zeta_k(R) = \zeta_k(S)$ for every integer $k \geq 2$, then $m = \ell$ and the multisets $\{n_1, \dots, n_m\}$ and $\{t_1, \dots, t_{\ell}\}$ agree.

Chakraborty and Patra introduced the invariant (1) in [1, Definition 2.1] and posed the conclusion of Theorem 1 in [1, Conjecture 4.11]. Their finite-product formula, chain-ring ideal description, and single-factor asymptotic provide the ingredients for the present proof [1, Corollary 3.2, Section 4.3, and Theorem 8.2]. The exact binomial complement is displayed in [1, proof of Theorem 8.5]. We include the reduction for completeness. The additional step is a separation argument for finite sums of logarithms, which recovers every length multiplicity at once. At the level of method, exact counting and analytic estimates of discrete probabilities also occur in Zeng's study of distinct cycle-length probabilities [2].

The source comparison is to the specified first version of [1]. A bounded literature search on 7 September 2026 identified no separate resolution; it does not establish exhaustive priority.

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2 Counting and separation

Lemma 2. *For positive integers n and integers $k \geq 2$, put*

$$q_n(k) = \frac{\binom{k+n-1}{n-1}}{(n+1)^k}, \quad f_n(k) = 1 - q_n(k). \quad (2)$$

A nonzero finite chain ring C of Loewy length n satisfies $\zeta_k(C) = f_n(k)$, and $0 < q_n(k), f_n(k) < 1$. For R as in Theorem 1,

$$\zeta_k(R) = \prod_{i=1}^m f_{n_i}(k). \quad (3)$$

Proof. Every ideal J of a finite chain ring is principal: the finite totally ordered family of principal ideals (x) , $x \in J$, has a largest member, and that member contains every element of J . There is a unique maximal ideal M , since any two maximal ideals are comparable. Write $M = (\pi)$. Elements outside M are units, so $1 - a\pi$ is a unit for every $a \in C$. Finiteness makes the powers (π^s) eventually stationary. If $(\pi^s) = (\pi^{s+1})$, then $\pi^s = a\pi^{s+1}$ for some a , and hence $(1 - a\pi)\pi^s = 0$. Thus $\pi^s = 0$. It follows that π is nilpotent and no two successive powers coincide before zero.

Every nonzero element is a unit times a power of π . Indeed, a nonunit can be factored by π , and repeating this operation must reach a unit before n steps, since $\pi^n = 0$. Consequently the complete ideal list is

$$C = M^0, M^1, \dots, M^{n-1}, M^n = (0), \quad M^a M^b = M^{\min(a+b, n)}.$$

The argument includes $M = 0$ and $n = 1$.

Choosing k ideals uniformly is therefore equivalent to choosing $a_1, \dots, a_k$ independently from $\{0, \dots, n\}$. Their product is nonzero exactly when $\sum_i a_i \leq n - 1$. Adjoining the slack coordinate $n - 1 - \sum_i a_i$ gives a bijection with weak compositions of $n - 1$ into $k + 1$ parts. There are $\binom{k+n-1}{n-1}$ such compositions; the individual upper bounds on the a_i are automatic. This proves (2). The tuple consisting only of C has nonzero product, and any tuple containing (0) has zero product, proving both strict inequalities.

For the product formula, every ideal J of $\prod_i R_i$ equals the product of its projected ideals J_i . To see the reverse inclusion, multiply an element of J by the i th coordinate idempotent to obtain any required J_i -element supported in that coordinate, and then sum over the coordinates. Under this bijection, ideal multiplication is componentwise: generators of a coordinate ideal product lift to products of elements supported in that coordinate. Uniform choice on this Cartesian product makes the coordinate choices independent. The product of the sampled ideals is zero exactly when it is zero in every coordinate, giving (3). $\square$

Lemma 3. *Let $(d_n)_{n \geq 1}$ be a nonzero finitely supported family of integers, and let $r = \min\{n : d_n \neq 0\}$. With the notation (2),*

$$\lim_{k \rightarrow \infty} \frac{-\sum_n d_n \log f_n(k)}{q_r(k)} = d_r, \quad (4)$$

where k ranges over integers.

Proof. For fixed n , the numerator of $q_n(k)$ is a polynomial of degree $n - 1$, so $q_n(k) \rightarrow 0$. If $n > r$, then

$$\frac{q_n(k)}{q_r(k)} = \frac{(r-1)!}{(n-1)!} \prod_{j=r}^{n-1} (k+j) \left(\frac{r+1}{n+1} \right)^k \rightarrow 0. \quad (5)$$

Here a fixed polynomial times ρ^k tends to zero for $0 < \rho < 1$: the ratio of successive terms of $k^a \rho^k$ tends to ρ and is eventually bounded above by $(1 + \rho)/2 < 1$.

For $0 < x < 1$, integration of $1/(1 - t)$ over $[0, x]$ gives

$$1 \leq h(x) := \frac{-\log(1 - x)}{x} \leq \frac{1}{1 - x}. \quad (6)$$

In particular $h(q_n(k)) \rightarrow 1$ for each fixed n . Therefore

$$\frac{-\sum_n d_n \log f_n(k)}{q_r(k)} = d_r h(q_r(k)) + \sum_{n>r} d_n h(q_n(k)) \frac{q_n(k)}{q_r(k)} \rightarrow d_r.$$

The sum is finite, and its indices and coefficients are fixed before taking the limit. $\square$

Proof of Theorem 1. Let a_n and b_n be the multiplicities of length n in R and S , and put $d_n = a_n - b_n$. These integer families have finite support. By (3) and positivity, the hypothesis implies

$$0 = \log \zeta_k(R) - \log \zeta_k(S) = \sum_n d_n \log f_n(k) \quad (k \geq 2).$$

If some d_n were nonzero, Lemma 3 would yield $0 = d_r \neq 0$ for the least such index r . Thus $a_n = b_n$ for every n , and summing the multiplicities gives $m = \ell$. $\square$

3 An explicit bound and the scope of recovery

Proposition 4. *For d and r as in Lemma 3, define*

$$E_d(k) = |d_r| \frac{q_r(k)}{1 - q_r(k)} + \sum_{n>r} |d_n| \frac{q_n(k)}{q_r(k)(1 - q_n(k))}. \quad (7)$$

Then

$$\left| \frac{-\sum_n d_n \log f_n(k)}{q_r(k)} - d_r \right| \leq E_d(k), \quad E_d(k) \rightarrow 0.$$

In particular, the rational inequality $E_d(k) < |d_r|$ certifies that $\sum_n d_n \log f_n(k)$ is nonzero and has sign $-\text{sgn}(d_r)$.

Proof. Equation (6) implies $|h(x) - 1| \leq x/(1 - x)$. Apply this to the r th term in the proof of Lemma 3, use $h(x) \leq 1/(1 - x)$ for the other terms, and take absolute values. Convergence follows from (5). The sign conclusion follows since $q_r(k) > 0$. $\square$

Thus each fixed pair of distinct length multisets admits a separation certificate using rational arithmetic alone. The bound depends on the pair; no parameter-independent finite sampling bound is asserted. In particular, a single value need not determine the lengths. The multisets $\{1, 1\}$ and $\{7\}$ both give $\zeta_2 = 9/16$, whereas their values at $k = 3$ are $49/64$ and $107/128$, respectively.

There is no restriction on factor counts, positive lengths, or repeated lengths in Theorem 1. Common lengths are cancelled through d_n before its least nonzero index is chosen. The proof does not require multiplicative independence of the integers $n+1$: all errors in replacing $-\log(1 - q_n)$ by q_n are controlled by (6). The invariant recovers lengths, without determining residue fields, characteristics, or ring isomorphism types; for example, all finite fields give $\zeta_k = 1 - 2^{-k}$. Nonprincipal rings and infinite products are outside the statement. Zero rings are excluded as chain factors. If the zero ring is admitted as an empty product, its value $\zeta_k = 1$ distinguishes it from every nonempty product considered here.

Computation and assistance. The proof is deductive and uses no finite computation to establish its unbounded conclusion. Supplementary Python computations with integers and exact fractions checked direct ideal counts for twelve small ring models at $k = 2, 3, 4$ and ten rational separation certificates. A separately written verifier checked the certificates and direct divisor-ideal counts for $\mathbb{Z}/m\mathbb{Z}$ with $m \in \{2, 3, 4, 8, 9, 12, 18, 24, 36\}$ at $k = 2, 3, 4, 5$. OpenAI Codex assisted with proof development, verification code, a separate model-context mathematical review, and manuscript preparation. The logarithmic proof route was already present in the initial research materials. The model-context review is not external peer review, and no proof-assistant formalization is claimed.

References

- [1] S. Chakraborty and S. K. Patra, *Ideal Zero-Product Probability in Finite Commutative Rings*, arXiv:2608.14618v1 (2026). <https://arxiv.org/abs/2608.14618v1>.
- [2] Z. Zeng, *The Distinct-Cycle-Length Probability is Strictly Decreasing after 30*, SSRN preprint (2026). <https://doi.org/10.2139/ssrn.7380519>.